Summary

Section 3-1: Addressing Overview

- Addressing methods operate at the data link, network, transport, and application layers of the OSI model.
- A MAC address is embedded on every NIC on the globe and is assumed to be unique to that NIC.
- An IP address is assigned to nearly every interface and can be used to find any computer in the world if the IP address is public on the Internet.
- A port is a number used by the transport layer to identify one application among several that might be running on a host.
- Every host on a network is assigned a unique character-based name called the FQDN (fully qualified domain name) consisting of a host name and a domain name.
- MAC addresses contain two parts, are 48 bits long, and are written as hexadecimal numbers separated by colons. The first 24 bits (six hex characters) are known as the OUI (Organizationally Unique Identifier), which identifies the NIC's manufacturer. The last 24 bits make up the extension identifier or device ID and identify the device itself.

Section 3-2: IP Addresses

- You can permanently assign a static IP address to a device, or you can configure the device to request and receive (or lease) a dynamic IP address from a DHCP server each time it connects to the network. A DHCP (Dynamic Host Configuration Protocol) server manages the dynamic distribution of IP addresses to devices on a network.
- A subnet mask indicates what portion of an IP address is the network portion, called the network ID or network address, and what part is the host portion, called the host ID or node ID.
- When using classful addressing, which is an older method of managing IP address ranges, the dividing line between the network and host portions is determined by the numerical range the IP address falls in. In contrast to classful addressing, classless addressing allows the dividing line between network and host portions to fall anywhere along the string of binary bits in an IP address. Moving this dividing line allows for segmenting networks within networks in a process called subnetting.
- Class A (1.x.y.z to 126.x.y.z), class B (128.0.x.y to 191.255.x.y), and class C (192.0.0.x to 223.255.255.x) addresses, for the most part, can be used to connect to and access Internet resources. Class D (begins with octets 224 through 239) and class E (begins with 240 through 254) IP addresses are not available for general use.
- To conserve its public IP addresses, a company can instead use private IP addresses for devices on its private networks—that is, devices that do not directly connect to the Internet but instead communicate through a representative device such as a router. These addresses were set aside for private use by IANA's RFC1918 (Request for Comment 1918) document, released in 1996.
- A broadcast message is read by every node on the network. A LAN, which consists of all the nodes a broadcast reaches, can be referred to as a broadcast domain. Routers don't forward broadcast messages, thus creating a boundary for a LAN.
- A DHCP scope, or DHCP pool, defines a range of IP addresses to be assigned to clients when they request an address. Other scope options include limiting the lease time and identifying the default gateway and DNS servers.
- When private devices need access to other networks or the Internet, a public-facing gateway (such as a router or firewall) substitutes the private IP addresses used by computers on the private network with its own public IP address. This process is called NAT (Network Address Translation). Besides requiring only a single public IP address for the entire private network, another advantage of NAT is security; the gateway hides the entire private network behind this one address.
- IPv6 offers a lot more public IP addresses than does IPv4 and also provides improvements to routing capabilities and speed of communication. In IPv6, a link, sometimes called the local link, is any LAN bounded by routers. Neighbors are two or more nodes on the same link.
- IPv6 supports three types of IP addresses, classified by how the address is used: unicast address (including global address, link local address, and loopback address), multicast address, and anycast address. IPv6 reduces network traffic by eliminating broadcasting.
- IPv6 addressing uses a process called SLAAC (stateless address autoconfiguration) so that a computer can autoconfigure its own link local IP address without the help of a DHCPv6 server. After generating an IPv6 address and confirming the address is unique on the network, the computer asks if a router on the network can provide configuration information. If a router responds with DHCP information, the computer uses whatever information this might be, such as the IP addresses of DNS servers or the network prefix.

Section 3-3: Ports and Sockets

- Whereas an IP address is used to find a computer, a port is used to find a process running on that computer. A socket consists of both a host's IP address and a process's TCP or UDP port, with a colon separating the two values.
- Well-known ports range from 0 to 1023 and are assigned by IANA to widely used and well-known utilities and applications. Registered ports range from 1024 to 49151 and can be used temporarily by processes for nonstandard assignments to increase security. Dynamic and private ports range from 49152 to 65535 and are open for use without restriction.
- Which port a protocol communicates over becomes especially relevant when configuring firewalls. A firewall works by blocking traffic on all ports except those that are specifically approved by the network admin.

Section 3-4: Domain Names and DNS

- Name resolution is the process of discovering the IP address of a host when its FQDN is known. An authoritative name server is the authority on computer names and their IP addresses for computers in their domains. The domains an organization is responsible for managing are called a DNS zone. DNS (Domain Name System) associates computer names with IP addresses. DNS namespace databases are stored on thousands of servers around the world, rather than being centralized on a single server or group of servers.
- The primary DNS server holds the authoritative DNS database for the organization's zones. When a secondary DNS server needs to update its database, it makes the request to the primary server for the update; this update process is called a zone transfer.
- A caching DNS server accesses public DNS data and caches the DNS information it collects. This server receives DNS queries from local network clients and works to resolve them by contacting other DNS servers for information. Caching DNS servers do not store zone files, which is why they must rely on their caches and resolution efforts.
- DNS name servers are organized in a global hierarchical structure. At the root level, 13 clusters of root DNS servers hold information used to locate the TLD (top-level domain) servers. These TLD servers hold information about the authoritative servers owned by various organizations.
- There are two types of DNS requests: recursive lookups, which demand a resolution or the answer that the information can't be found, and iterative lookups, which do not demand resolution and only provide information if the server already has it.
- Namespace databases are stored in DNS zone files, which are simple text files consisting of resource records that each store specific kinds of information about the zone. These records consist of fields specific to the kind of information that record type should hold, such as the zone name, class (these days, that's always Internet class), and record type. Other fields in each record vary according to the purpose of that record.

Section 3-5: Troubleshooting Address Problems

- The ping (Packet Internet Groper) utility is used to verify that TCP/IP is installed, bound to the NIC, configured correctly, and communicating with the network. The ping utility starts by sending out a signal called an echo request to another computer, which is simply a request for a response. The other computer then responds to the request in the form of an echo reply. The protocol used by the echo request and echo reply is ICMP (Internet Control Message Protocol), a lightweight protocol used to carry error messages and information about a network.
- The `ipconfig` command shows current TCP/IP addressing and domain name information on a Windows computer. You can also use `ipconfig` to change some of these settings.
- On UNIX and Linux systems, use the ip utility to view and manage TCP/IP settings. As with ipconfig on Windows systems, you can use ip to view and modify TCP/IP settings and to release and renew the DHCP configuration. Any `ip` commands that change the state of a link require elevated privileges. Any changes to link configurations are not saved by default—they will be lost upon the next restart unless the configuration changes are added to a startup script or edits are made to underlying configuration files.
- Similar to ip, use the ifconfig utility to view and manage TCP/IP settings on UNIX and Linux systems. The ifconfig utility has been deprecated in many Linux distributions, including Ubuntu. This means it's not installed by default, and users are intended to use the `ip` command instead. However, you can install it by running the command `sudo apt install net-tools`.
- The hostname utility offers very basic commands used to display a device's host name, either in Windows, UNIX, or Linux system. In Windows, hostname has no additional parameters. In UNIX or Linux, hostname offers a few more options including the ability to change the computer's name.

- The nslookup (name space lookup) utility allows you to query the DNS database from any computer on the network and find the host name of a device by specifying its IP address, or vice versa. This is useful for verifying that a host is configured correctly or for troubleshooting DNS resolution problems.
- The dig (domain information groper) utility is available on Linux and macOS and provides more detailed domain information than nslookup. Use dig to query DNS nameservers for information about host addresses and other DNS records.

Key Terms

For definitions of key terms, check the Glossary.

A (address) record
AAAA (address) record
address pool exhaustion
anycast address
APIPA (Automatic Private IP Addressing)
authoritative name server
broadcast
broadcast domain
caching DNS server
canonical name
CIDR (Classless Interdomain Routing) notation
classful addressing
classless addressing
CNAME (canonical name) record
default gateway
device ID
DHCP (Dynamic Host Configuration Protocol)
DHCP reservation
DHCP scope
DHCP scope exhaustion
dig (domain information groper)
DNS zone
domain name
dual stacked
dynamic IP address
EUI-64 (Extended Unique Identifier-64)
exclusion range
forward lookup
forwarding DNS server
FQDN (fully qualified domain name)
gateway
global address
host ID
host name
hostname
hosts file
ICMP (Internet Control Message Protocol)
ifconfig
interface
interface ID
ip
IP reservation
IPAM (IP address management)
ipconfig
IPv4 (Internet Protocol version 4)
IPv6 (Internet Protocol version 6)
iterative lookup
LDAP (Lightweight Directory Access Protocol)
LDAPS (Lightweight Directory Access Protocol over SSL)
lease time
link
link local address
localhost
loopback address
MAC address table
MAC reservation
multicast
multicast address
multicast flooding
MX (mail exchange) record
name resolution
NAT (Network Address Translation)
NAT64 (Network Address Translation 64)
neighbor
network ID
NS (name server) record
nslookup (name server lookup)
NTP (Network Time Protocol)
NTS (Network Time Security)
octet
OUI (Organizationally Unique Identifier)
PAT (Port Address Translation)
ping (Packet Internet Groper)
primary DNS server
private IP address
process
PTP (Precision Time Protocol)
PTR (pointer) record
public IP address
recursive lookup
registered port
resource record
reverse lookup
RFC1918 (Request for Comment 1918)
root DNS server
scope option
secondary DNS server
session
SLAAC (stateless address autoconfiguration)
SMB (Server Message Block)
SOA (start of authority) record
socket
static IP address
stratum
subnet
subnet ID
subnet mask
subnetting
syslog (system log)
TFTP (Trivial File Transfer Protocol)
TLD (top-level domain)
TTL (time to live)
tunneling
TXT (text) record
unicast address
URL (uniform resource locator)
well-known port
zone transfer

Acronyms Checklist

The acronyms in Table 3-12 are listed in the CompTIA Network+ objectives and could appear on the Network+ exam. This means that exam questions might use any of these acronyms in context so that you must know the meaning of the acronym in order to answer the question correctly. Make sure you're familiar with what each acronym stands for and the general concept of the term itself. All these acronyms are used in context in this chapter.

Table 3-12 Chapter 3 acronyms

Acronym	Spelled out
A	address
APIPA	Automatic Private Internet Protocol Addressing
CIDR	Classless Inter-Domain Routing
CNAME	canonical name
DHCP	Dynamic Host Configuration Protocol
DNS	Domain Name System
GUI	graphical user interface
IPAM	Internet Protocol address management
LDAP	Lightweight Directory Access Protocol
LDAPS	Lightweight Directory Access Protocol over SSL
MX	mail exchange
NAT	Network Address Translation
NS	name server
NTP	Network Time Protocol
NTS	Network Time Security
PAT	Port Address Translation
PTP	Precision Time Protocol
PTR	pointer
SLAAC	stateless address autoconfiguration
SMB	Server Message Block
SMTPS	Simple Mail Transfer Protocol Secure
SOA	start of authority
TTL	time to live
TXT	text
URL	uniform resource locator

Review Questions

1. What type of device does a computer turn to when attempting to connect with a host with a known IP address on another network?
   a. Default gateway
   b. DNS server
   c. Root server
   d. DHCP server
2. What decimal number corresponds to the binary number 11111111?
   a. 255
   b. 256
   c. 127
   d. 11,111,111
3. Suppose you send data to the 11111111 11111111 11111111 11111111 IP address on an IPv4 network. To which device(s) are you transmitting?
   a. All devices on the Internet
   b. All devices on your local network
   c. The one device that is configured with this IP address
   d. No devices
4. When your computer first joins an IPv6 LAN, what is the prefix of the IPv6 address the computer first configures for itself?
   a. FF00::/8
   b. ::1/128
   c. 2000::/3
   d. FE80::/64

5. If you are connected to a network that uses DHCP, and you need to terminate your Windows workstation's DHCP lease, which command would you use?

   a. `ipconfig /release`
   b. `ipconfig /renew`
   c. `ifconfig /release`
   d. `ifconfig /renew`

6. Which DNS server offers the most current resolution to a DNS query?

   a. TLD DNS server
   b. Root DNS server
   c. Caching DNS server
   d. Primary DNS server

7. Which of the following IP addresses could be configured by a host computer for itself?

   a. 192.168.0.5
   b. 192.169.254.1
   c. 169.254.192.1
   d. 10.0.0.0

8. Which type of DNS record identifies an email server?

   a. AAAA record
   b. CNAME record
   c. MX record
   d. PTR record

9. You issue a transmission from your workstation to the following socket on your LAN: 10.25.80.22:53. Assuming your network uses standard port designations, what application layer protocol handles your transmission?

   a. SSH
   b. SMTP
   c. HTTP
   d. DNS

10. You have just set up a new wireless network at your house, and you want to determine whether your Linux laptop has connected to it and obtained a valid IP address. What command will give you the information you need? Choose TWO.

    a. `ipconfig /all`
    b. `ip address show`
    c. `ifconfig -a`
    d. `hostname`

Scenario-Based Questions

Question 3-1

You have just brought online a new secondary DNS server and notice your network-monitoring software reports a significant increase in network traffic. Which two hosts on your network are likely to be causing the increased traffic and why?

a. The caching and primary DNS servers, because the caching server is requesting zone transfers from the primary server.
b. The secondary and primary DNS servers, because the secondary server is requesting zone transfers from the primary server.
c. The root and primary DNS servers, because the primary server is requesting zone transfers from the root server.
d. The web server and primary DNS server, because the web server is requesting zone transfers from the primary DNS server.

Question 3-2

You are the network manager for a computer training center that allows students to bring their own laptops to class for learning and taking notes. Students need access to the Internet, so you have configured your network's DHCP server to issue IP addresses automatically. Which DHCP option should you modify to make sure you are not wasting addresses used by students who have left for the day?

a. Address pool
b. DNS server address
c. Lease time
d. IP reservation

Question 3-3

You have decided to use SNAT and PAT on your small office network. At minimum, how many IP addresses must you obtain from your ISP for all five clients in your office to be able to access servers on the Internet?

a. 0
b. 1
c. 5
d. 10

Question 3-4

While troubleshooting a network connection problem for a coworker, you discover the computer is querying a nonexistent DNS server. What command-line utility can you use to assign the correct DNS server IP address?

a. `nslookup`
b. `hostname`
c. `ping`
d. `ifconfig`

Question 3-5

When running a scan on your computer, you find that a session has been established with a host at the address 208.85.40.44:443. Which application layer protocol is in use for this session? What command-line utility might you use to determine the domain name of the other computer?

a. Telnet; nslookup
b. IMAP4; hostname
c. HTTPS; nslookup
d. RDP; hostname

Hands-On Projects

Note 3-19 | Websites, applications, public cloud platforms, and related account options change often. While the instructions given in these projects were accurate at the time of writing, you might need to adjust the steps or options according to later changes.

Note to Instructors and Students: A rubric is provided for evaluating student performance on these projects. Please see Appendix D.

Project 3-1: Create a NAT Translation Table Entry

Estimated time: 20 minutes
Objective 2.1: Explain characteristics of routing technologies.
Resources:
- No special resources are required for this project

Description:
Your corporation hosts a website at the static public IP address 92.110.30.123. A router directs this traffic to a web server at the private IP address 192.168.11.100. However, the web server needs a hardware upgrade and will be down for two days. Your network administrator has asked you to configure the router so that

requests to the IP address 92.110.30.123 are redirected to the backup server for the website, which has the private IP address 192.168.11.110. The router's inside Ethernet interface uses IP address 192.168.11.254, and its outside interface uses the IP address 92.110.30.65.

Respond to the following questions about the new static route you'll be creating:

1. What is the router's outside interface IP address?
2. What is the router's inside interface IP address?
3. What is the website's public IP address?
4. What is the private IP address of the backup web server?

With this information in hand, use the example given in Figure 3-16 earlier in the chapter as a template to create the NAT translation table entries for the address translation. For the subnet masks, use the default subnet mask for a Class C IP address license. Include appropriate comment lines in your table. **Take a screenshot of your NAT translation table**; submit this visual with your responses to this project's questions.

Project 3-2: Manage a DNS Cache

Estimated time: 10 minutes
Objective 3.4: Given a scenario, implement IPv4 and IPv6 network services.
Resources:
- Windows computer with administrative access
- Internet access

Description:
You have learned that clients as well as name servers store DNS information to associate names with IP addresses. In this project, you display the contents of a local DNS cache, clear it, and display it again after performing some DNS lookups. Then you change DNS servers and display the DNS cache once again. Complete the following steps:

1. To display the DNS cache, open an elevated PowerShell or Command Prompt window and enter the following command: `ipconfig /displaydns`
2. If this computer has been used to resolve host names with IP addresses—for example, if it has been used to retrieve email or browse the web—a list of locally cached resource records appears. Explore the file to see what kinds of records have been saved, using the scroll bar if necessary. What is the most common record type in this list?
3. Clear the DNS cache with this command: `ipconfig /flushdns`
   The operating system confirms that the DNS resolver cache has been flushed. One circumstance in which you might want to empty a client's DNS cache is if the client needs to reach a host whose IP address has changed (for example, a website whose server was moved to a different hosting company). If the DNS information is locally cached, the client will continue to look for the host at the old location. Clearing the cache allows the client to retrieve the new IP address for the host.
4. Display the DNS cache again with the command: `ipconfig /displaydns`
5. Open a browser window and navigate to three websites you have not recently visited, such as **howstuffworks.com**, **nautil.us**, and **mapcrunch.com**.
6. Return to the PowerShell or Command Prompt window and display the DNS cache containing the new list of resource records. If the list is still empty or doesn't show records for the sites you visited, you might be using a browser that performs its own DNS processing rather than communicating with the Windows service. In this case, ping three more

websites in your PowerShell or Command Prompt window so that Windows performs the DNS processing. For example, you can enter `ping archive.org` to force a name resolution in Windows. Then display the DNS cache again.

7. **Take a screenshot** of one of the new records that was collected in response to your browser or ping activity and shows the URL of the target website; submit this visual with your responses to this project's questions.

Exam Tip | The CompTIA Network+ exam includes two types of questions: performance-based questions (called PBQs) and multiple-choice questions. You're already familiar with multiple-choice questions. PBQs give you a scenario of some kind and require you to interact with tools such as a network diagram, user interface, or command-line interface to complete tasks related to the scenario. Common performance-based questions on the Network+ exam require you to know how to manage DNS from a CLI using `ipconfig`. You might also see a multiple-choice question that covers similar information.

Capstone Projects

Note 3-20 | Websites, applications, public cloud platforms, and related account options change often. While the instructions given in these projects were accurate at the time of writing, you might need to adjust the steps or options according to later changes.

Note to Instructors and Students: A rubric is provided for evaluating student performance on these projects. Please see Appendix D.

Capstone Project 3-1: Set Up an Ubuntu Server in a VM

Estimated time: 45 minutes
Objective 5.5: Given a scenario, use the appropriate tool or protocol to solve networking issues.
Resources:
- Access to the same computer used to complete Capstone Project 1-1
- Internet access
- If desired, instructor can provide Ubuntu Server image file

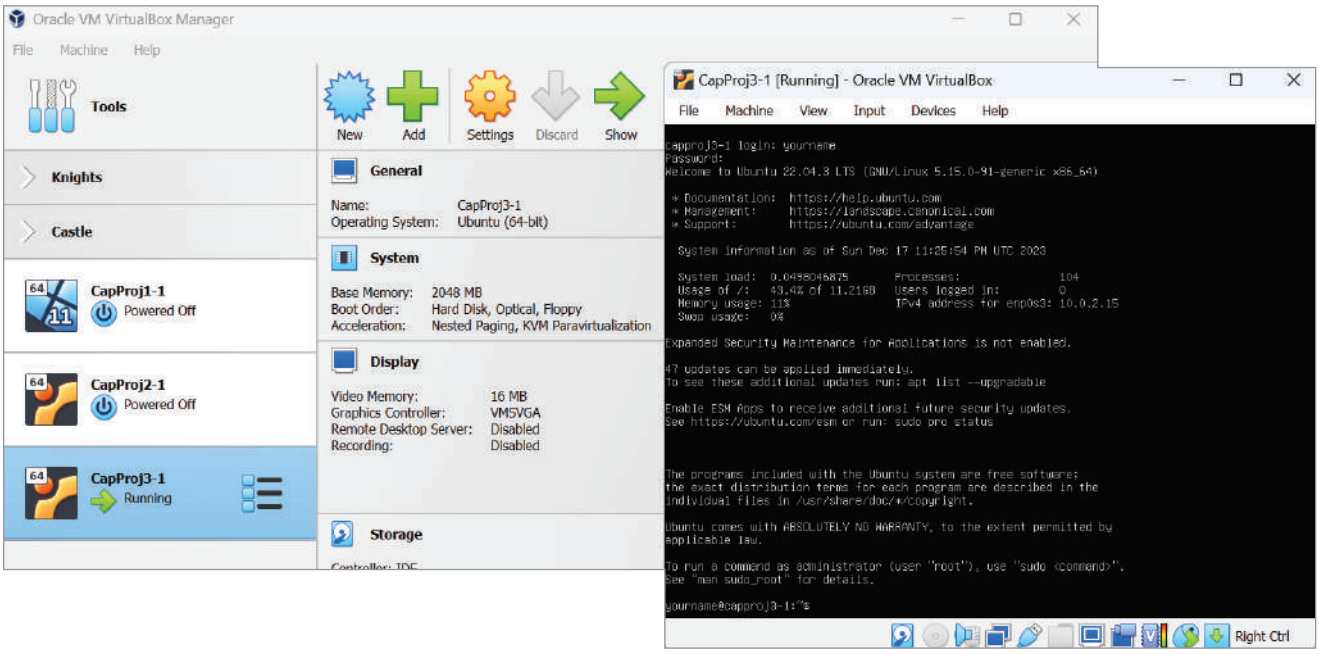
Description:
In the Chapter 1 Capstone Project, you created a Windows virtual machine using Oracle VirtualBox. In Capstone Project 2-1, you added a second VM, this one running Ubuntu Desktop. In this Capstone Project, you create a third VM and install Ubuntu Server in the VM. You also learn how to use some Linux commands. Using the same computer that you used in Capstone Project 1-1, complete the following steps:

1. Go to **ubuntu.com/server** and download the Ubuntu Server OS to your hard drive using **Option 1 - Manual server installation**. If you're given the choice of multiple versions, choose the latest version. The file that downloads is an ISO file.

Note 3-21 | Ubuntu Server is only available as a 64-bit OS. To install a 64-bit guest OS in a VM, the host OS must also be 64-bit.

2. Open the Oracle VM VirtualBox Manager. Refer back to the directions in the Chapter 1 Capstone Project if needed to create the VM and install the Ubuntu Server OS. Give the VM an informative name. Be sure to record your Ubuntu host name, username, and password in your Bitwarden vault. You can choose an updated installer, if available. When given the option, decline to install any extra software bundled with the OS other than standard system utilities. Accept all other default settings. When prompted, reboot the VM.
3. After you restart the VM, Ubuntu Server launches, which does not have a GUI. Enter your username and password to log in. The shell prompt should now be displayed (Figure 3-41). What is your current shell prompt?

Figure 3-41 Ubuntu Server is installed in a VirtualBox VM



Source: Oracle Corporation; Canonical Group Limited

4. Practice using Ubuntu Server by entering, in order, each of the commands listed in Table 3-13. As you do so, you'll examine the directory structure, create a new directory, and create a blank file in the new directory.

Table 3-13 Practice Linux commands in Ubuntu Server

Step	Command	Description
1	`pwd`	Displays the full path to the current directory. When you first log in to a system, that directory is /home/*username*.
2	`mkdir mydir`	Creates a directory named mydir. The new directory is created in the current directory. You must have permission to edit the current directory.
3	`dir`	Lists files and directories in the current directory. In Linux, a directory is treated more like a file than a Windows folder.
4	`cd mydir`	Goes to the directory you just created in the /home/*username* directory. What is your new shell prompt?
5	`touch myfile`	Creates a blank file named myfile in the current directory.

Step	Command	Description
6	`ls`	Similar to `dir`, lists current directory contents.
7	`cd ..`	Moves up one level in the directory tree. **Take a screenshot** of the commands you've entered so far and their output; submit this visual with your responses to this project's questions.
8	`cd /etc`	Changes directory to the /etc directory, where text files are kept for configuring installed programs. What is your new shell prompt?
9	`ls`	Examines the contents of the /etc directory.
10	`ls -l`	Lists the contents of the /etc directory again but with additional information.
11	`cd /home`	Changes directory to the /home directory.
12	`ping 127.0.0.1`	Pings the loopback address. Pinging continues until you stop it by pressing Ctrl+C.
13	**Ctrl+C**	Cancels a command or process; use it to recover after entering a wrong command or to stop a command that requires you manually halt the output.
14	`ifconfig`	Displays basic TCP/IP information and network information, including the MAC address of the NIC. However, the ifconfig utility is no longer installed on Ubuntu Server by default.
15	`sudo apt update` and enter your password	Updates the apt database in the local OS.
16	`sudo apt install net-tools`	Installs the net-tools suite, which includes the ifconfig utility.
17	`ifconfig`	Now that it is installed, the ifconfig utility should display basic TCP/IP information and network information, including the MAC address of the NIC.
18	`ip addr`	Displays IP and MAC addresses. Notice the difference in output for `ifconfig` compared to `ip addr`. What is your VM's IPv4 address? Note that Linux calls this address inet. (Be careful you don't answer with the loopback address! Most likely, the active network connection is on the enp0s3 or eth0 interface.)
19	`ip help`	Displays available objects and options for the `ip` command.
20	`dig google.com`	Performs a DNS lookup on the *google.com* domain name. How many A records are listed?
21	`df`	Displays the amount of free space on your hard drive. In this case, the VM is reporting statistics for its virtual hard drive.
22	`exit`	Logs out; the login shell prompt appears, where you can log in again. Enter your username and password to log in again.
23	`sudo poweroff`	Elevates privilege to shut down the VM. You'll need to enter your password, and then the system shuts down.

5. Add the new VM's information to your VMclients page in your wiki. On the Virtualization:VMclients page, click **Edit** at the bottom of the page and add the new VM to your list. Include the chapter number, hypervisor used, VM computer name, and VM operating system. Also note any additional information that you might find helpful when you return to this VM in the future, such as how to view the Linux Manual or how to shut down the system gracefully. When you're finished, click **Save**.
6. **Take a screenshot** of the edited wiki page; submit this visual with your responses to this project's questions.

Capstone Project 3-2: Build a MAC Address Table in Packet Tracer

Estimated time: 45 minutes

Objective 5.5: Given a scenario, use the appropriate tool or protocol to solve networking issues.

Resources:

- Computer with Cisco Packet Tracer installed
- Internet access

Description:

In Capstone Project 2-2, you installed Packet Tracer and practiced interacting with the user interface. Earlier in this chapter, you learned about MAC address tables that switches use to track which device is connected to each of a switch's ports. In this Capstone Project, you build a small network in Packet Tracer and observe changes to a switch's MAC address table. Complete the following steps:

1. Open Cisco Packet Tracer and, if necessary, sign in with your Skills for All account.
2. In the Devices pane, click **Network Devices** category and then click **Switches**. Add a **PT-Switch** to your workspace. Give the switch a moment to boot.
3. Click **Switch0** to open its configuration window. Click the **CLI** tab. This takes you to the CLI for this switch where you can enter commands to interact with the switch. While Packet Tracer offers some options for configuring devices through their GUIs, the tasks in this project can only be completed from the CLI.
4. Click at the bottom of the IOS Command Line Interface pane in the empty space below "Press RETURN to get started!" Press **Enter** to activate the CLI. By default, you begin in user EXEC command mode, which has the lowest level of privileges in a Cisco device. You can see what mode you're in by looking at the prompt: user EXEC mode shows the prompt `Switch>`. To enter privileged EXEC mode, enter `enable`. The prompt changes to `Switch#`.
5. Now that you're in privileged EXEC mode, you can check the switch's current MAC address table. Enter `show mac-address-table`. What entries are listed?
6. Minimize the switch's configuration window and return to your workspace. Click the **End Devices** category in the Devices pane, and then add two **PCs** to your workspace.
7. Click **PC0**. In PC0's configuration window, click the **Desktop** tab and then click **IP Configuration**. In this project and most Packet Tracer projects in this course, you'll set static IP addresses. Enter the following information and then close the configuration window (the information saves automatically):

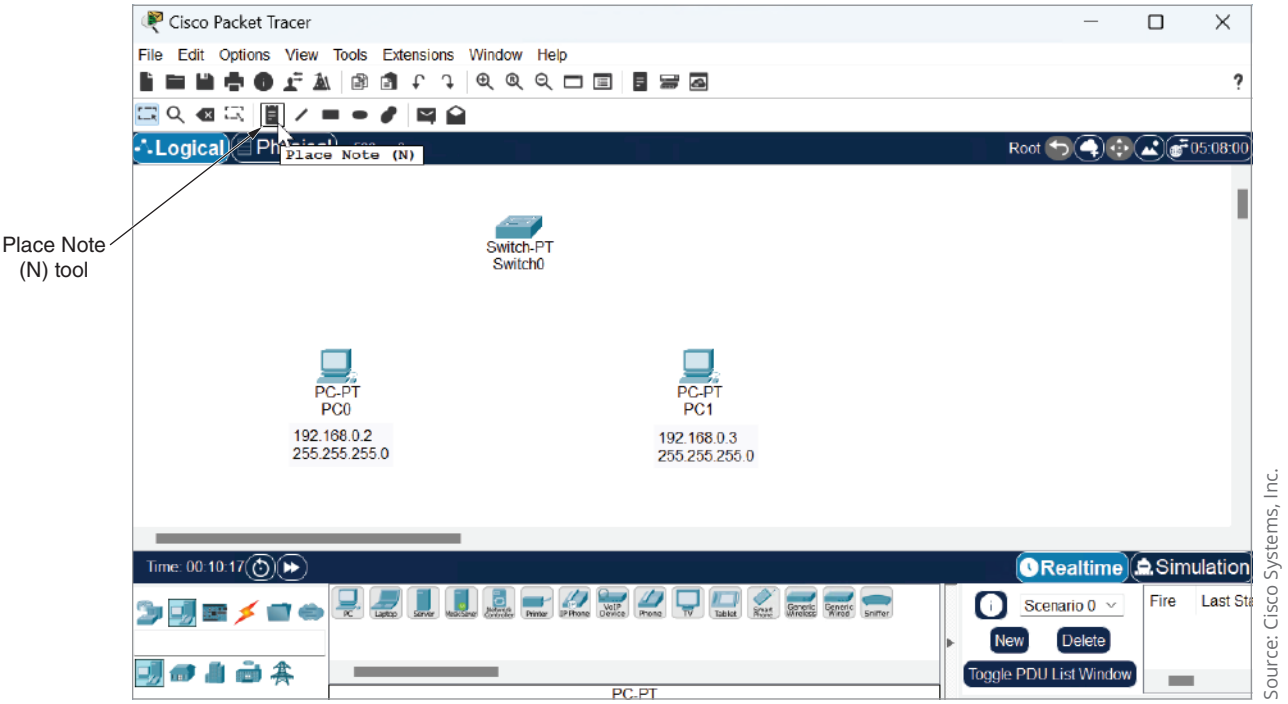
   IP address: **192.168.0.2**
   Subnet mask: **255.255.255.0**

8. Repeat Step 7 for PC1 and enter the following information for PC1:

   IP address: **192.168.0.3**
   Subnet mask: **255.255.255.0**

9. It's important to get in the habit of keeping good documentation as you work. In the toolbar above your workspace, click the **Place Note (N)** tool. Click under each PC and document that device's IP address and subnet mask (Figure 3-42).
10. Now you're ready to connect your PCs to your switch. In the Devices pane, click **Connections** and then click the **Copper Straight-Through** cable, which is a thick, straight, black line. Click **PC0** and select its **FastEthernet0** interface. Then click **Switch0** and select its **FastEthernet0/1** interface. Repeat this process for PC1, connecting PC1's **FastEthernet0** interface to Switch0's **FastEthernet1/1** interface. Wait for all indicator lights to turn into green triangles.

Figure 3-42 Use notes to document your network configurations



Source: Cisco Systems, Inc.

11. Access Switch0's CLI again. Click at the bottom of the CLI pane and press **Enter**. Check Switch0's MAC address table. What entries are listed?

Recall that the switch must see traffic crossing its interfaces to collect MAC addresses for connected devices. To generate traffic, run a ping from PC0 to PC1. Complete the following steps:

12. Click **PC0** and, if necessary, click the **Desktop** tab. Click **Command Prompt**. At the `C:\>` prompt, enter `ping 192.168.0.3` and wait for ping to complete.
13. Return to Switch0's CLI and check its MAC address table again. **Take a screenshot** of the output; submit this visual with your responses to this project's questions.

A switch can only see network traffic that crosses its interfaces. It's possible for traffic from multiple devices to enter a switch at a single switch port. In this case, the switch will record multiple MAC addresses for a single interface. Complete the following steps:

14. Add a second **PT-Switch** and a third **PC**.
15. Configure PC2 with the following information and create a note to document this configuration:

    IP address: **192.168.0.4**
    Subnet mask: **255.255.255.0**

16. In the **Connections** group, click the **Fiber** cable, which is the solid, straight, orange line. Click **Switch0** and select its **FastEthernet4/1** interface. Then click **Switch1** and select its **FastEthernet4/1** interface.
17. Use a **Copper Straight-Through** cable to connect PC2's **FastEthernet0** interface to Switch1's **FastEthernet0/1** interface. Wait for all indicator lights to turn into green triangles.
18. Check Switch0's MAC address table again. What entries are listed? Given this information, which connected device is Switch0 currently aware of?
19. Sending a ping between PC2 and PC0 will inform Switch0 of three devices' MAC addresses. Which devices do you expect Switch0 to know about after the ping?

20. From PC2, ping PC0. Did the ping work?
21. Return to Switch0's CLI and check its MAC address table again. **Take a screenshot** of the output; submit this visual with your responses to this project's questions.
22. Examine the three devices listed in Switch0's MAC address table and respond to the following questions:
    a. How many devices is Switch0 currently aware of?
    b. Which device is connected to Switch0's Fa0/1 interface? How can you confirm which device matches this MAC address?
    c. Which two devices communicated across Switch0's Fa4/1 interface?
23. Currently, PC1 is not showing in Switch0's MAC address table. What can you do to make Switch0 aware of PC1?
24. You do not need to save this Packet Tracer network for future projects. However, before closing the network, take some notes in your Wikidot website about your work in this project, commands that you learned, and new insights you have about how Packet Tracer works.

Note to Instructors: A Packet Tracer solution file is provided for each Packet Tracer project through the Instructors website.

Solutions to Self-Check Questions

Section 3-1: Addressing Overview

1. What numbering system do humans use to read and write MAC addresses?
   Answer: d. Hexadecimal
   Explanation: A MAC address is 48 bits, written as 12 **hexadecimal** characters separated by colons, as in 00:60:8C:00:54:99. A hexadecimal number (also called a hex number) is a number written in the base-16 number system, which uses the 16 numerals 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, and F. IPv4 addresses are written as four decimal numbers called octets, for example, 92.106.50.200. Each octet, when written in binary, consists of exactly 8 bits. A binary number is a number written in the base-2 number system, which uses only the numerals 0 and 1.

2. What Windows command outputs a computer's MAC address?
   Answer: c. `ipconfig /all`
   Explanation: In a PowerShell or Command Prompt window, you can enter **`ipconfig /all`** to identify your NIC's MAC address. The `ipconfig` command without the `/all` parameter will not show the MAC address. The `ping` command is used to check connectivity between two devices. Oui-lookup is not a standard Windows command.

Section 3-2: IP Addresses

3. What command shows you a computer's TCP/IP configuration?
   Answer: b. `ipconfig`
   Explanation: You can find TCP/IP configuration information and more using the **`ipconfig`** command. The `ping` command is used to check connectivity between two devices. The `ssh` command creates an encrypted channel or tunnel between two computers. Nmap is a simple and popular command-line utility used to detect network devices.

4. Which of the following IPv4 addresses is a public IP address?
   Answer: d. 64.233.177.189
   Explanation: IP addresses within the ranges 10.0.0.0 through 10.255.255.255, 172.16.0.0 through 172.31.255.255, and 192.168.0.0 through 192.168.255.255 are RFC1918, or private, IP addresses. The address **64.233.177.189** is a public IP address and goes to one of Google's servers.

5. Which IPv6 prefix can be routed on the Internet?
Answer: a. 2000::/3
Explanation: Global IPv6 addresses that can be routed on the Internet often begin with the prefix **2000::/3**, although other prefixes are being released. Link local addresses begin with FE80::/10. The loopback address is written ::1/128. Many unique local unicast addresses begin with FC00::/7.

Section 3-3: Ports and Sockets

6. When hosting a secure email server for access from the Internet, which port should be open on the corporate firewall?
Answer: d. 587
Explanation: SMTP (Simple Mail Transfer Protocol) is used to send email messages to and between email servers. The old port for SMTP is 25, but encrypted transmissions to an SMTP server use port **587**. Port 110 is the nonsecure port for POP3, which is another email protocol. Port 443 is the secure port for encrypted HTTP.

7. Which port should be open so you can remote into the corporate office's Linux Server from a branch office?
Answer: a. 22
Explanation: In Linux operating systems, SSH (Secure Shell) provides secure, encrypted remote access and works over port **22**. Port 3389 serves RDP (Remote Desktop Protocol) on Windows systems. Port 23 supports the unsecure Telnet. Port 1433 supports communications with Microsoft SQL Server.

Section 3-4: Domain Names and DNS

8. When your computer requests a DNS lookup, which DNS server holds the most reliable information for that DNS record?
Answer: c. Primary DNS server
Explanation: The **primary DNS server** is the authoritative name server for the organization, which holds the authoritative DNS database for the organization's zones. The caching DNS server and forwarding DNS server might contain needed DNS information in their cache, but this information could potentially be outdated. Root DNS servers play a role in DNS resolution by holding information used to locate the TLD (top-level domain) servers. These TLD servers hold information about the authoritative servers owned by various organizations.

9. Which DNS record type is listed first in a zone file?
Answer: d. SOA
Explanation: An **SOA (start of authority)** record is listed at the beginning of the zone file and gives important information about the zone, such as a contact email address, when the zone was last updated, how long the zone information is valid until it should be refreshed, and necessary information for how to perform a zone transfer. A (address) records, AAAA (address for IPv6) records, and CNAME (canonical name) records are placed later in the zone file and provide mapping information.

10. Which DNS record type is used to find an FQDN from a given IP address?
Answer: c. PTR
Explanation: A **PTR (pointer)** record is used for reverse lookups to provide a host name when you know an IP address. A (address) records find IPv4 addresses when the host name is known. CNAME (canonical name) records hold alternative names for a host. An MX (mail exchange) record identifies an email server.

Section 3-5: Troubleshooting Address Problems

11. What protocol does ping use?
    Answer: b. ICMP
    Explanation: The protocol used by the ping echo request and echo reply is **ICMP (Internet Control Message Protocol)**, a lightweight protocol used to carry error messages and information about a network. HTTP is used for communications with web servers. DHCP is used to dynamically assign IP addresses on a network. FTP is a file transfer protocol.

12. Which command disables a computer's connection to the network?
    Answer: c. `ipconfig /release`
    Explanation: The `ipconfig /release` command releases the IP address when dynamic IP addressing is being used. Releasing the IP address effectively disables the computer's communications with the network until a new IP address is assigned. The `sudo ip link set enp0s3 up` command brings the enp0s3 interface to an "up" state. The `sudo hostname testVM` command sets a new host name on a device. The `ping localhost` command pings the loopback address on a device.

13. What problem will `ping google.com` check for that `ping 8.8.8.8` will not?
    Answer: d. Name resolution issues
    Explanation: Pinging a host using its host name can verify you have Internet access *and* functioning **name resolution (DNS) services** because the host name must first be resolved before the host can be located and pinged. Pinging any device outside your network will check default gateway configuration, even if you ping its IP address instead of its host name. Pinging a target's host name or IP address will not determine whether you're using the correct DNS server. DHCP scope exhaustion is not determined with the `ping` command.